Lecture 20

The Weierstrass function

Objectives. *A meromorphic function can repeat in two independent complex directions. We construct such a function by a convergent sum over a lattice. Its poles determine a cubic differential equation. The equation relates the lattice description to a genus-one curve. The analytic construction and differential equation are proved; the final geometric identification is a sketch.*

By the end of this lecture you will be able to:

- Justify the subtracted lattice series by an explicit convergence bound.
- Prove evenness and periodicity of the Weierstrass function.
- Derive its differential equation and identify its half-period values.
- Locate the point at infinity in the torus–cubic sketch.

20.1 Lattices and elliptic functions

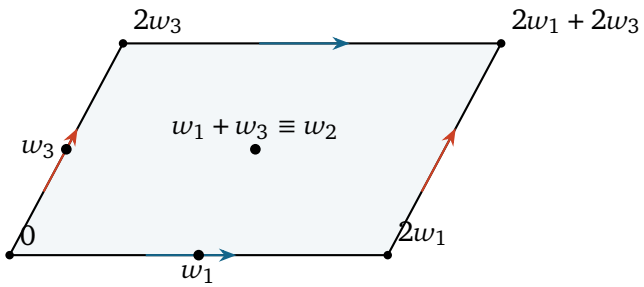
Choose nonzero complex numbers w_1 and w_3 with $\operatorname{Im}(w_3/w_1) > 0$. The *lattice* they generate is

$$\Lambda = 2w_1\mathbb{Z} + 2w_3\mathbb{Z}, \qquad w_2 = -w_1 - w_3.$$

The generators $2w_1$ and $2w_3$ are full periods; the numbers w_j are half-periods. A lattice point is denoted by ℓ, not by the spectral variable λ. Source half-periods written ω_j become w_j here, since the collection reserves ω for the symplectic form.

The closed *fundamental parallelogram* P_Λ has vertices 0, $2w_1$, $2w_1 + 2w_3$ and $2w_3$. Its translates cover the plane; edge points have more than one representative. A meromorphic function f on $\mathbb{C}$ is *elliptic for* Λ if $f(z + \ell) = f(z)$ for every $\ell \in \Lambda$. It suffices to verify this for the two generators.

The diagram marks the three nonzero half-period classes. Two points represent the same class if their difference belongs to Λ.



Opposite edges are identified in the indicated directions. The point $w_1 + w_3$ represents $w_2 = -w_1 - w_3$ because their difference is in Λ.

20.2 Liouville theorems for elliptic functions

The *order* of a nonconstant elliptic function is the number of its poles in one cell, counted with multiplicities and with identified edges counted only once. For a nonconstant function a shifted cell can avoid zeros and poles on its boundary: there are finitely many such points modulo Λ, by isolation of zeros and poles and compactness of the cell.

Theorem 20.1. *A holomorphic elliptic function is constant. The sum of the residues of an elliptic function in one cell is zero. A nonconstant elliptic function has equally many zeros and poles in one cell, counted with multiplicity. In particular it cannot have exactly one simple pole modulo Λ.*

Proof. A holomorphic elliptic function is bounded on the compact cell and hence on the plane. Liouville's theorem, Theorem 10.4, makes it constant.

Choose a translated cell with no boundary pole. The integrals of f over opposite edges cancel because those edges differ by a period and have opposite boundary orientations. The residue theorem (Theorem 11.3) gives a zero sum of residues. A lone simple pole would have nonzero residue, which is impossible.

For nonconstant f, choose the boundary to avoid its zeros as well. The function f'/f is elliptic, so its opposite-edge integrals also cancel. The argument principle, Theorem 11.5, identifies its integral divided by $2\pi\mathrm{i}$ with the number of zeros minus the number of poles. Thus that difference is zero. ∎

For example, an elliptic function with one double pole and no other poles has two zeros counted with multiplicity. The same holds for $f - c$ for every $c \in \mathbb{C}$, since subtracting a constant does not change the poles. The construction below realises precisely this pole pattern.

20.3 The convergent lattice series

The real-linear map $(m, n) \mapsto 2m\omega_1 + 2n\omega_3$ is invertible on $\mathbb{R}^2$. Its inverse is bounded on the unit circle, so some constant $c > 0$ gives

$$|2m\omega_1 + 2n\omega_3| \geq c \max(|m|, |n|).$$

There are $8r$ integer pairs with $\max(|m|, |n|) = r$, for $r \geq 1$. Consequently $\sum_{\ell \in \Lambda \setminus \{0\}} |\ell|^{-p} < \infty$ whenever $p > 2$, by comparison with $\sum_{r \geq 1} 8r(cr)^{-p}$.

Define the *Weierstrass function* by the subtracted series

$$\wp(z) = \frac{1}{z^2} + \sum_{\ell \in \Lambda \setminus \{0\}} \left(\frac{1}{(z - \ell)^2} - \frac{1}{\ell^2} \right), \qquad z \notin \Lambda. \tag{20.1}$$

The subtraction is needed for absolute convergence over a two-dimensional lattice. This construction follows Stein–Shakarchi II, pp. 267–270. [1] Here *normal convergence* means that the sum of the supremum norms of the summands on each compact set off the poles is finite.

[1] Their full periods $1, \tau$ correspond to $2\omega_1, 2\omega_3$ after the change of variable $z/(2\omega_1)$. Their lattice summand uses $z + \ell$; replacing ℓ by $-\ell$ gives our expression.

Theorem 20.2. *The series converges absolutely and locally uniformly off Λ. It defines a meromorphic function with a double pole of leading term $(z-\ell)^{-2}$ at every lattice point and no other poles. Its derivative is*

$$\wp'(z) = -2\sum_{\ell\in\Lambda}\frac{1}{(z-\ell)^3}. \tag{20.2}$$

This derivative series is absolutely and locally uniformly convergent off the poles.

Proof. Fix $R > 0$. For $|z| \le R$ and $|\ell| > 2R$, the exact identity and bound are

$$\frac{1}{(z-\ell)^2} - \frac{1}{\ell^2} = \frac{2z\ell - z^2}{\ell^2(z-\ell)^2}, \qquad \left|\frac{2z\ell - z^2}{\ell^2(z-\ell)^2}\right| \le 10R|\ell|^{-3}.$$

The preceding lattice estimate and the M-test prove uniform absolute convergence of the tail. On a compact set avoiding Λ, the remaining finitely many summands are holomorphic. The locally uniform limit of holomorphic functions is holomorphic, as proved in § 10.6. Near a lattice point isolate its single singular summand; the other terms have a holomorphic sum. This gives the stated double pole.

The differentiated tail has bound $16|\ell|^{-3}$ on the same disk. Its series is uniformly convergent there. Termwise differentiation, justified by the derivative-convergence result in § 10.6, gives (20.2), including the term at $\ell = 0$. ∎

20.4 Parity and periods

Proposition 20.3. *The function $\wp$ is even and elliptic for Λ.*

Proof. Replace ℓ by $-\ell$ in the absolutely convergent series (20.1). This proves $\wp(-z) = \wp(z)$. The derivative series is absolutely convergent too. Replacing ℓ by $\ell + 2w_j$ proves $\wp'(z + 2w_j) = \wp'(z)$, for $j = 1, 3$. Hence $\wp(z + 2w_j) - \wp(z)$ has zero derivative off the lattice. This complement is connected, so the difference is a constant. At $z = -w_j$ neither term is a pole, and evenness makes their difference zero. Thus both generators are periods. ∎

The same construction also records scaling. If the lattice is shown as an argument, direct substitution in its absolutely convergent series gives

$$\wp(cz \mid c\Lambda) = c^{-2}\wp(z \mid \Lambda), \qquad c \in \mathbb{C}\setminus\{0\}. \tag{20.3}$$

For the square lattice $\Lambda = 2\mathbb{Z} + 2\mathrm{i}\mathbb{Z}$, multiplication by i permutes the lattice. Scaling therefore gives $\wp(\mathrm{i}z) = -\wp(z)$. The same reindexing in the sum of sixth reciprocal powers changes its sign, so that sum is zero. This example will make $g_3 = 0$ below.

20.5 The cubic differential equation

Define the lattice invariants by absolutely convergent sums

$$g_2 = 60\sum_{\ell\neq 0}\ell^{-4}, \qquad g_3 = 140\sum_{\ell\neq 0}\ell^{-6}.$$

All omitted lattice indices belong to Λ. A prime on a sum omits zero. Expanding $(z-\ell)^{-2} = \ell^{-2}\sum_{n\ge 0}(n+1)(z/\ell)^n$ for $|z|$ smaller than the shortest nonzero lattice length gives

$$\wp(z) = z^{-2} + az^2 + bz^4 + O(z^6), \qquad a = \frac{g_2}{20}, \quad b = \frac{g_3}{28}. \tag{20.4}$$

Absolute convergence on a smaller disk permits summation in either order. Odd powers cancel in pairs $\ell, -\ell$; the constant was subtracted. The coefficients of z^2 and z^4 are respectively $3\sum' \ell^{-4}$ and $5\sum' \ell^{-6}$.

Theorem 20.4. *The Weierstrass function satisfies*

$$\wp'(z)^2 = 4\wp(z)^3 - g_2\wp(z) - g_3. \tag{20.5}$$

Proof. Form $G = \wp'^2 - 4\wp^3 + g_2\wp + g_3$, an elliptic function. At zero, the Laurent expansion gives

$$G(z) = (g_2 - 20a)z^{-2} + (g_3 - 28b) + O(z^2) = O(z^2).$$

The z^{-6} terms cancel and there are no other negative powers. Periodicity gives the same removable singularity at every lattice point. Thus G is holomorphic and elliptic, hence constant by Theorem 20.1. Its limit zero at the origin makes this constant zero. ∎

Set $e_j = \wp(w_j)$ for $j = 1, 2, 3$. These labels follow DLMF 23.3.9 and Lawden, not the index order in Stein–Shakarchi II.[2]

Proposition 20.5. *The three numbers e_j are distinct, and*

$$\wp'^2 = 4(\wp - e_1)(\wp - e_2)(\wp - e_3), \qquad e_1 + e_2 + e_3 = 0.$$

Proof. The three half-periods are distinct nonzero classes modulo Λ. Since $2w_j \in \Lambda$, oddness and periodicity of $\wp'$ give $\wp'(w_j) = \wp'(-w_j) = -\wp'(w_j)$, so each is a zero. The derivative has one triple pole in a cell. By Theorem 20.1, it has exactly three zeros counted with multiplicity. The three listed zeros are therefore simple and exhaustive. If $e_j = e_m$ for two different classes, the function $\wp - e_j$ would have a zero of order at least two at each, since its derivative vanishes there. This contradicts its order two. Thus the values are distinct. Equation (20.5) makes them roots of $4X^3 - g_2X - g_3$. Factoring this cubic proves the first formula; its missing quadratic coefficient proves their sum is zero. ∎

For the square lattice, $g_3 = 0$ as above. The cubic therefore has zero as one root and the remaining roots are opposite. This follows from the lattice symmetry, without choosing an ordering of real roots for a general complex lattice.

20.6 The torus and the cubic

The quotient $\mathbb{C}/\Lambda$ is the set of equivalence classes under $z \sim z + \ell$. Its coordinates come from small disks that do not meet their nontrivial lattice translates. Opposite edges of a cell are identified, so the quotient is a compact *torus*, a genus-one surface. This is the same topological gluing interpretation used in Lecture 19.

[2] On their normalised lattice, printed p. 271, our e_1, e_2, e_3 equal their e_1, e_3, e_2, respectively, multiplied by $(2w_1)^{-2}$.

Projective coordinates $[X : Y : Z]$ are nonzero triples in $\mathbb{C}^3$, identified under multiplication by the same nonzero scalar. The projective cubic is

$$Y^2Z = 4X^3 - g_2XZ^2 - g_3Z^3.$$

On $Z \neq 0$ it is the affine curve $y^2 = 4x^3 - g_2x - g_3$. Its three finite branch values are distinct by Proposition 20.5. Its only point with $Z = 0$ is $[0 : 1 : 0]$.

The geometric identification is

$$[z] \longmapsto [\wp(z) : \wp'(z) : 1], \qquad [0] \longmapsto [0 : 1 : 0] \qquad \text{(geometric sketch)}.$$

The ODE puts the image on the cubic. The assignment is independent of the representative because both functions are periodic. At zero, multiply the triple by z^3; the Laurent expansion gives the limit $[z + O(z^5) : -2 + O(z^4) : z^3] \to [0 : 1 : 0]$. For a generic value of $\wp$, its two preimages are z and $-z$, since it has order two. Their derivative values are opposite, so the pair $(\wp, \wp')$ distinguishes them. At a half-period they coincide in the quotient. These observations explain the identification. The conclusion that it is a holomorphic isomorphism, meaning a one-to-one holomorphic map with holomorphic inverse, remains a sketch; see Dubrovin, pp. 72–73.[3]

The path computation of Lecture 19 can also be read in the lattice coordinate. Along a lifted path avoiding poles and zeros of $\wp'$, set $x = \wp(z)$ and $y = \wp'(z)$. The chain rule gives

$$\frac{\mathrm{d}x}{y} = \frac{\wp'(z)\,\mathrm{d}z}{\wp'(z)} = \mathrm{d}z.$$

Choose a path from a generic z_0 to $z_0 + 2\omega_j$ avoiding those isolated points, for $j = 1, 3$. Its image on the cubic is closed, by periodicity. Its differential integral is

$$\oint_{\gamma_j} \frac{\mathrm{d}x}{y} = \int_{z_0}^{z_0+2\omega_j} \mathrm{d}z = 2\omega_j, \qquad j = 1, 3.$$

Thus the two lattice directions give explicit period paths. This calculation uses the chain rule on the path; it does not supply the omitted global isomorphism proof or a general period-lattice theorem.

A rectangular lattice also has an elementary limiting form. For fixed $a > 0$, $\omega_1 = a$ and $\omega_3 = \mathrm{i}b$, DLMF 23.12.1 gives

$$\wp(z) \longrightarrow \frac{\pi^2}{4a^2}\left(\frac{1}{\sin^2(\pi z/(2a))} - \frac{1}{3}\right), \qquad b \to \infty, \qquad z \notin 2a\mathbb{Z} \quad \text{(cited)}.$$

To obtain the rotated limit explicitly, write $\Lambda_{a,b} = 2a\mathbb{Z} + 2\mathrm{i}b\mathbb{Z}$ for $a, b > 0$. Rotation sends this lattice to $\Lambda_{b,a}$, up to the sign of one generator. The scaling identity gives

$$\wp(z \mid \Lambda_{a,b}) = -\wp(-\mathrm{i}z \mid \Lambda_{b,a}).$$

Apply the cited trigonometric limit to the right-hand side as $a \to \infty$, keeping b fixed. With $u = \pi z/(2b)$, away from the poles, the elementary identity

$$\sin(-\mathrm{i}u) = -\mathrm{i}\sinh u, \qquad -\left(\frac{1}{\sin^2(-\mathrm{i}u)} - \frac{1}{3}\right) = \frac{1}{\sinh^2 u} + \frac{1}{3}$$

[3] His full generators $2\omega, 2\omega'$ become $2\omega_1, 2\omega_3$. We use the directly proved cubic signs in (20.5), also fixed by DLMF 23.3.10.

accounts for both signs in the resulting expression.

As a consequence of this cited limit and (20.3), fixing the imaginary half-period ib and sending the real one to infinity gives the hyperbolic limit

$$\frac{\pi^2}{4b^2}\left(\frac{1}{\sinh^2(\pi z/(2b))}+\frac{1}{3}\right), \qquad z \notin 2\mathrm{i}b\mathbb{Z}.$$

For the rotation by –i, the rotated half-periods may be relabelled as b and ia, since changing a generator's sign leaves the lattice unchanged. Their ratio has positive imaginary part, as required. Scaling both half-periods to infinity in a fixed shape gives z^{-2} by the Laurent expansion and (20.3). The cited trigonometric limit and its rotated consequence explain the potentials in integrability; their general degeneration theory is not proved here.

Summary. The subtracted lattice sum converges normally and defines an even, doubly periodic meromorphic function. Its Laurent coefficients determine the cubic ODE. The torus–cubic interpretation retains its geometric sketch status.

- Absolute convergence justifies reindexing the derivative series to prove periods.
- The half-period values are the distinct roots of $4X^3 - g_2X - g_3$.
- A lattice point corresponds to the cubic's point $[0 : 1 : 0]$ at infinity.

References

Stein–Shakarchi, *Complex Analysis*, Ch. 9, pp. 262–274. Lawden, *Elliptic Functions and Applications*, pp. 149–158, confirms the half-period convention; its theta-first construction is not used here. Dubrovin, *Integrable Systems and Riemann Surfaces*, pp. 72–73, describes the torus–cubic map. DLMF 23.2, 23.3.9–10 and 23.12.1 fix the normalisations and the cited limit.